

Leyton Kiel Mueller

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EDUCATION

University of Illinois Urbana-Champaign
Computer Engineering, B.S.

Expected May 2028
GPA: 3.76/4.00

Relevant Coursework: Digital Signal Processing, Digital Systems Lab, Computer Systems Engineering, Analog Signal Processing, DS & Algorithms, Computer Systems & Programming

Organizations, Awards, & Honors: James Scholar, Grainger First Year Scholar, Oakley Scholarship Recipient, AIM High Grant Nominee, Engineering Visionary Scholarship Recipient, Dean's List, Illini EV Concept Embedded Team

WORK EXPERIENCE

Incoming Lead Systems Engineer Intern
Advanced Micro Devices (AMD)

Fall 2026
Austin, TX

- ACPI power state validation, stress and benchmark automation, and system-level root-cause analysis for desktop CPU platforms

FPGA R&D Intern
Sandia National Laboratories

May 2026 - August 2026
Livermore, CA

- Architected SoC with native SD interface, AXI interconnect FIFO, & MicroBlaze soft processor on Xilinx FPGA
- Achieved 5x throughput (10→50 MHz) through parallelization and native high-speed 4-bit SD mode
- Implemented production-grade protocol logic in VHDL with sub-nanosecond latency optimization, validated timing constraints via static timing analysis and place-and-route
- Designed AXI FIFO datapath with cross-clock domain isolation, integrating asynchronous data buffers for deterministic processor-peripheral synchronization with external telemetry instrumentation

FPGA Intern
Institut Central Lille

May 2025 - Jul 2025
Villeneuve-d'Ascq, France

- Developed an FPGA-based spectroscopy platform, replicating waveform generators and oscilloscopes' functions on a Xilinx Zynq SoC, replacing equipment with a compact, high-speed signal acquisition system
- Achieved sustained, real-time streaming up to 250 MSPS with zero packet losses, leveraging custom Vivado modules for high-throughput DMA and adaptive trigger logic
- Optimized hardware/software co-design by developing C++ drivers for direct memory-mapped I/O, reducing end-to-end latency and enabling sub-microsecond control and measurement loops
- Designed automated Python-based test infrastructure and live diagnostics, verifying system reliability and performance across billions of samples in continuous operation

PROJECTS

Hardware Ethernet Stack | SystemVerilog, Python

- Designed RTL Ethernet stack for bidirectional frame exchange between an Artix-7 FPGA and host laptop through a network switch
- Built 100 Mbps Ethernet MAC on Artix-7 with MDIO controller, MII nibble serialization, preamble/SFD framing, interframe gap enforcement, and CRC-32 FCS validated by residue check
- Implemented receive path with SFD alignment, MAC address filtering, and FCS validation driving an RTL ARP responder, verifying frame structure and link behavior against Wireshark and Scapy on the host

SKILLS

Programming: C, C++, SystemVerilog, VHDL, Python, RISC-V Assembly, TCL, Bash

FPGA Architecture & Protocols: FIFO, BRAM, SPI, PCIe, TCP/UDP, Ethernet, AXI interconnect, DMA

Developer Tools: Git, VS Code, Vivado, Vitis, JIRA, Linux CLI